Lahore Urban Estates Analytics Warehouse

Business Requirements Document

DL3004 Data Warehousing and Business Intelligence — Final Project — Phase 1 · Version 2.1
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This document states what the warehouse must deliver and how each requirement will be accepted. It translates stakeholder needs into business, data, quality, reporting and non-functional requirements, and traces every business requirement to the questions and facts that satisfy it.

1. Purpose and Audience

The BRD is the agreement between LUE management and the project team. Its readers are the six stakeholder groups, the course instructor and the team members who will build Phase 2 and Phase 3.

2. Business Background

LUE manages about 1,800 units in 40 buildings across Lahore, Karachi, Islamabad and Faisalabad for private owners. Its data is spread across a property management system, Excel registers, settlement files, a tenant maintenance app and leasing spreadsheets. Cross-building questions take days to answer, and history is lost whenever a record is updated.

3. Stakeholders and Their Needs

CEO

Needs: Portfolio health, risk
Example question: Which buildings underperform on collection and NOI?
Report page: Owner portfolio

Finance head

Needs: Collections, arrears, cash flow
Example question: What is this month's collection rate and arrears ageing?
Report page: Collections

Operations head

Needs: Vacancy, maintenance performance
Example question: Which vendors breach SLA most?
Report page: Maintenance

Building managers

Needs: Their own building
Example question: Who is overdue in my building? Which requests are aged?
Report page: Collections, Maintenance (filtered)

Leasing manager

Needs: Renewals, vacancy, incentives
Example question: Do incentives shorten vacancy?
Report page: Occupancy and leasing

Owner relations lead

Needs: Owner reporting
Example question: What is each owner's NOI this quarter?
Report page: Owner portfolio

4. Business Requirements

Priority uses MoSCoW: *Must* requirements are needed for acceptance; *Should* requirements are planned but can be reduced if time runs short.

BR-01. Report the monthly collection rate by building, city and region, with subtotals

Stakeholder: Finance head
Priority: Must
Answered by: Q1

BR-02. Rank buildings by arrears ratio and show overdue balances by ageing bucket

Stakeholder: Finance head, CEO
Priority: Must
Answered by: Q2, Q3

BR-03. Identify tenants with recurring late payment and compare segments at the time of payment

Stakeholder: Finance head
Priority: Must
Answered by: Q4

BR-04. Compare payment channels on lateness and settlement speed

Stakeholder: Finance head
Priority: Should
Answered by: Q5

BR-05. Show daily occupancy and vacancy trends by city and unit type

Stakeholder: Operations head
Priority: Must
Answered by: Q6

BR-06. Measure vacancy duration by unit type

Stakeholder: Leasing manager
Priority: Must
Answered by: Q7

BR-07. Report renewal rate and tenant loss at renewal by building

Stakeholder: Leasing manager
Priority: Must
Answered by: Q8

BR-08. Measure maintenance speed, SLA breaches and cost by category, vendor and building

Stakeholder: Operations head
Priority: Must
Answered by: Q9, Q10, Q11

BR-09. Report net operating income per owner portfolio

Stakeholder: CEO, owner relations lead
Priority: Must
Answered by: Q12

BR-10. Measure whether vacancy incentives shorten vacancy

Stakeholder: Leasing manager
Priority: Should
Answered by: Q13

BR-11. Relate maintenance demand to weather

Stakeholder: Operations head
Priority: Should
Answered by: Q14

BR-12. Reproduce any past monthly report exactly, even after master data changes

Stakeholder: All
Priority: Must
Answered by: Q4, Q9

Acceptance. A requirement is met when its questions are answered by SQL on the warehouse, with output and a one-line interpretation, and the same figures appear in the Power BI report.

5. KPI Definitions

Every KPI is computed from additive parts (sums and counts) and divided at the end. Averaging per-row ratios gives the wrong answer when rows have different weights (Lab 05 Task 6.3).

Collection rate

Formula: SUM(payment_amount allocated to lines due in the period) / SUM(net_amount of lines due in the period)
Facts used: fact_rent_invoice_line, fact_payment
Question: Q1

Arrears ratio

Formula: SUM(net_amount − allocated payments) for lines past due / SUM(net_amount billed)
Facts used: fact_rent_invoice_line, fact_payment
Question: Q2

Arrears ageing

Formula: Open balance grouped by days past due: 0–30, 31–60, 61–90, 90+
Facts used: fact_rent_invoice_line, fact_payment
Question: Q3

DSO (rent)

Formula: Open receivable at period end / net_amount billed in the period × days in the period
Facts used: fact_rent_invoice_line, fact_payment
Question: Q2

Occupancy rate

Formula: SUM(is_occupied) / COUNT(unit-days)
Facts used: fact_unit_occupancy_daily
Question: Q6

Average vacancy duration

Formula: Total vacant days / number of completed vacancy spells
Facts used: fact_unit_occupancy_daily
Question: Q7

Renewal rate

Formula: COUNT(renewed events) / COUNT(leases whose end_date falls in the period)
Facts used: fact_lease_event, dim_lease
Question: Q8

Average days to resolve

Formula: SUM(days_to_resolve) / COUNT(resolved requests)
Facts used: fact_maintenance_request
Question: Q9

SLA breach rate

Formula: SUM(sla_breached_flag) / COUNT(resolved requests)
Facts used: fact_maintenance_request
Question: Q10

Maintenance cost per unit-month

Formula: SUM(actual_cost) / (number of units × number of months)
Facts used: fact_maintenance_request, dim_unit
Question: Q11

Net operating income per owner

Formula: Rent collected − SUM(actual_cost) − LUE management fee
Facts used: fact_payment, fact_maintenance_request
Question: Q12

Incentive effect

Formula: Average vacancy spell length with an offer − without an offer
Facts used: fact_vacancy_incentive_coverage, fact_unit_occupancy_daily
Question: Q13

6. Data Requirements

Sources: six systems: PostgreSQL (S1), Excel (S2), CSV (S3 and S6), JSON (S4) and a REST API (S5)
Granularity: the lowest available level: invoice line, payment allocation, unit-day, maintenance request, lease event and offer-day
History: three years; SCD Type 2 on tenant type and sector, unit type and furnishing, vendor rating and SLA policy
Ownership: the owner on each fact row is resolved from the ownership register at the event date
Calendar: calendar and fiscal (July–June) hierarchies, with public holiday and Ramadan flags
Late-arriving data: inferred dimension members with deferred update; late facts loaded at their true date
Refresh: daily batch, with incremental loads by watermark per source

7. Data Quality Requirements

DQ-01. NULL check

Rule: Business keys, amounts and dates must be present
On failure: Quarantine; non-critical text defaults to 'Unknown'

DQ-02. Data type check

Rule: Dates parse in every building's format; amounts are numeric
On failure: Quarantine

DQ-03. Duplicate check

Rule: No repeated payment_reference or invoice number + line number in a batch
On failure: Keep the first, quarantine the rest

DQ-04. Business rule check

Rule: net = billed + tax − discount; lease end after start; resolved after raised; payment > 0 unless a reversal
On failure: Quarantine

DQ-05. Referential check

Rule: Every invoice's lease, unit and tenant exist
On failure: Missing tenant → inferred member; missing unit → quarantine

DQ-06. Range check

Rule: Dates between 2023-07-01 and the run date; amounts within plausible bounds
On failure: Quarantine

DQ-07. Reconciliation check

Rule: Source totals equal warehouse totals (rows, net_amount, payments per channel per day)
On failure: Run marked Warning; reported on the pipeline page

Every rejected row is stored in lue_audit.quarantine with its source system, source table, batch_id, check name, reason and timestamp. The generator's defect answer key is used to prove that each check works.

8. Reporting Requirements

Collections

Audience: Finance head, building managers
Main visuals: collection rate trend, ageing buckets, overdue tenants
Requirements: BR-01 to BR-04

Occupancy and leasing

Audience: Operations head, leasing manager
Main visuals: occupancy by city and unit type, vacancy duration, renewal rate, incentive effect
Requirements: BR-05 to BR-07, BR-10

Maintenance

Audience: Operations head
Main visuals: SLA breaches by vendor, days to resolve, rain vs leak requests
Requirements: BR-08, BR-11

Owner portfolio

Audience: CEO, owner relations lead
Main visuals: NOI per owner, maintenance cost per unit
Requirements: BR-09

Pipeline health

Audience: Project team, instructor
Main visuals: run log, rejects by reason, reconciliation status
Requirements: NFR-02

9. Non-Functional Requirements

- **NFR-01 Reproducibility.** One command rebuilds sources, warehouse, marts and checks from the seeded generator.
- **NFR-02 Auditability.** Every run logs rows read, inserted, updated and rejected per step; every rejected row is kept with its reason.
- **NFR-03 Lineage.** Every row carries source_system, batch_id and loaded_at; lue_audit.lineage maps each table to its sources.

- **NFR-04 Performance.** Three benchmark queries are measured with EXPLAIN ANALYZE before and after partitioning and indexing.
- **NFR-05 Usability.** Power BI pages load from Gold marts; every subtotal row in SQL output is labelled, never a bare NULL.
- **NFR-06 Privacy.** No real personal data; names, phones and companies are synthetic.
- **NFR-07 Documentation.** Grain sentences, table and column names are identical across all documents and the dictionary.
- **NFR-08 Refresh.** Daily batch; the last 35 days of the occupancy snapshot are re-derived to absorb late-arriving data.

10. Assumptions and Constraints

- LUE and all data are fictional; the generator is seeded, so every run produces identical data.
- A unit has one owner at a time; ownership can change on a dated transfer recorded in the S2 register.
- A lease has one primary tenant and one unit; a renewal creates a new lease linked to the previous one.
- Rent and service charges are invoiced monthly; payments may be partial and split across invoice lines.
- LUE's management fee is assumed to be 8% of rent collected (a parameter, used in Q12).
- Fiscal year runs July to June; all amounts are in Pakistani Rupees (PKR).
- The warehouse refreshes once a day in batch; intraday latency is not required.
- Co-ownership of one unit by several owners is out of scope (it would need a bridge table).
- Only the brief's free stack is used: PostgreSQL, pgAdmin, Python/pandas, Power BI Desktop, draw.io/Mermaid and GitHub.

11. Out of Scope

- Real or personal data, live integration, streaming and prediction
- Accounting-grade ledger and tax reporting
- Co-ownership of a unit by several owners

12. Requirements Traceability

- **BR-01 to BR-04** → Q1–Q5 → fact_rent_invoice_line, fact_payment → *Collections*
- **BR-05, BR-06** → Q6, Q7 → fact_unit_occupancy_daily → *Occupancy and leasing*
- **BR-07** → Q8 → fact_lease_event → *Occupancy and leasing*
- **BR-08** → Q9–Q11 → fact_maintenance_request → *Maintenance*
- **BR-09** → Q12 → fact_payment, fact_maintenance_request → *Owner portfolio*
- **BR-10** → Q13 → fact_vacancy_incentive_coverage, fact_unit_occupancy_daily → *Occupancy and leasing*
- **BR-11** → Q14 → fact_maintenance_request with dim_weather → *Maintenance*
- **BR-12** → Q4, Q9 → all facts through SCD Type 2 dimensions → *All pages*

Read each line as: requirement → questions → fact tables → report page.